

Lecture 7

Evan Drake

Recall Boolean Logic

True | False

- ◆ Booleans Basis?

- ◆ Boolean Gates?

Not, And, Or, XOr, NAnd

- ◆ Why?

“If you give someone Fortran, he has Fortran. If you give someone Lisp, he has any language he pleases.”

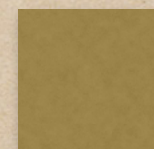
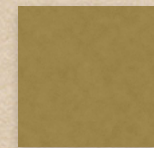
-Guy Steele

Control Flow



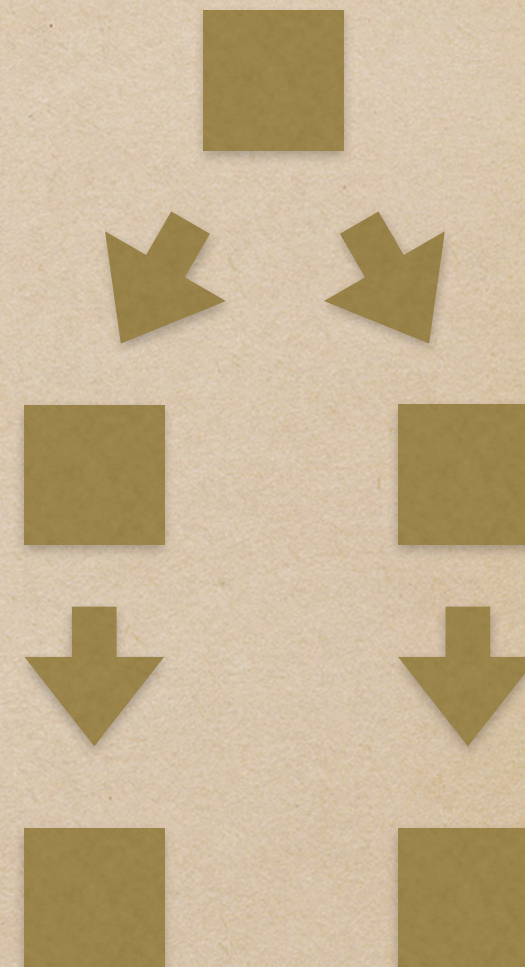
Control Flow

- ◆ Linear Programming?
- ◆ Can we improve?



Control Flow

- ◆ Dynamic invocation?
- ◆ “Forked” execution path



Control Flow

```
1 x = 5
2 y = 4
3 def f(x,y):
4     return x + y
5
6 sum = f(x,y)
```

1,2



3,4



6

Notice Linearity

Control Flow

- ◆ If-then-else
- ◆ Match

If-then-else

If it is **raining** then I will have an **umbrella**

If it is **cold** then I will take a **jacket**

If <boolean> then <action> else <action>

If-then-else

```
1 if <boolean>:  
2     <true-action>  
3 else:  
4     <false-action>
```

```
1 x = 5  
2 y = 4  
3 if x > y:  
4     print("x is greater")  
5 else:  
6     print("x is lesser")
```


Match

- ◆ Type dependent
- ◆ Structure dependent
- ◆ Useful for large Deltas

Also known as a **switch case**

“Anybody can write complicated code. It takes
genius to write simple code.”

-Walter Bright

Control Flow

```
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4     return x + y
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6 sum = f(x,y)
```

1,2



3,4

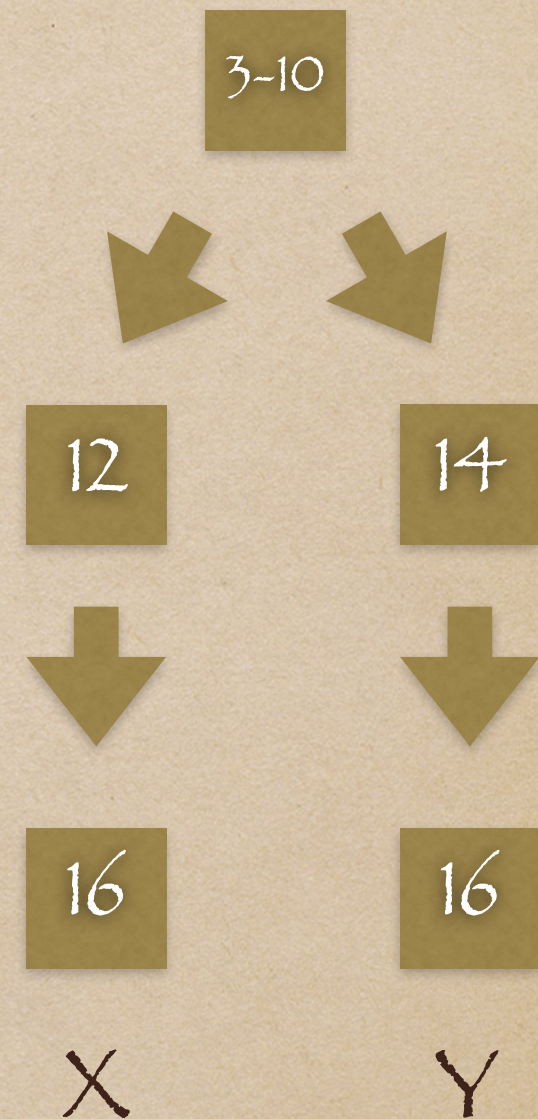


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Notice Linearity

Control Flow

```
1 import random
2
3 x = 5
4 y = 4
5 c = random.randint(0,10)
6
7 def f(x,y):
8     return x + y
9
10 def maybe(x,y,c):
11     if c:
12         return x
13     else:
14         return y
15
16 output = maybe(x,y, f(x,c) == 10)
```



Live Coding